



Integrated digital control room with human-in-the-loop for effective SMR operation

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ABSTRACT

This study introduces an integrated Digital Control Room (DCR) for improved operation performance of Small Modular Reactor (SMR). The Integrated DCR framework encompasses human performance monitoring, semantic reasoning, and adaptive controls into a single system with the considerations of human factors within the context of intelligent automation. The proposed design includes Human Performance Management System (HPMS), Human Performance Semantic Network (HPSN), Human Monitoring System (HMS), Human Simulation (HSIM), Control Room Controller (CRC), and External Environment Interface (EEI). The interactions among these modules provide an effective closed loop for effective and optimum SMR operation. The integrated monitoring system enables real-time assessment of operator performance metrics including stress, workload, and situational awareness, and link these with plant and environmental conditions. Operational scenarios are modeled using a co-simulation environment and integrated with SMR simulator for Integral Pressurized Water Reactor (IPWR) from IAEA, to assess operator states and their correlation with environmental changes and required adaptive control room configurations.

1. Introduction

As worldwide energy demand grows quickly, and there is a pressing need to transition to sustainable low carbon energy sources, advanced nuclear power technologies have again received considerable attention as trustworthy methods of meeting long-term electricity and heat demand. Small Modular Reactors (SMRs) represent a new class of technological innovation in the nuclear power sector. SMRs are factory-built compact reactors with individual capacities typically less than 300 MW (e) that can be deployed in modules, possess improved passive safety features, and offer scalable economics (International Atomic Energy Agency, 2023a; International Atomic Energy Agency, 2023b). Due to their size characteristics, SMRs can be manufactured in a standardized way and installed in less time than larger nuclear reactors. In addition, modular SMRs can serve hybrid energy systems, have load-following capability, operate safely with some increments of renewable energy and hybrid energy systems, and provide energy reliability while improving grid stability.

Although the technological advancements of SMRs represent a

significant step forward in the nuclear engineering arena, there is a corresponding need to update the human-system interface in concert with the development of the plant's digital and autonomous systems to support the evolution of the control room's operational capabilities. Traditional main control rooms were engineered in an analog instrumentation paradigm of minimal automation, extensive manual operation, and rigid ergonomics. These systems are increasingly inconsistent with contemporary reactors' operational complexity and digital nature. As nuclear plants develop toward more automated and data-based organizational structures, the operator's responsibilities will no longer simply entail direct manual control but rather be within a supervisory and educated decision-making role. This change brings about new forms of cognitive workload, new forms of situational awareness challenges, and new forms of human-machine mismatches that can impact safety and performance.

To respond to the emerging challenges, the Integrated Digital Control Room (IDCR) project is focused on the creation of a human-centered integrated control environment for SMRs enhanced with human performance and situational awareness, along with dynamic interaction.

Abbreviations: DCR, Digital Control Room; IDCR, Integrated Digital Control Room; SMR, Small Modular Reactor; HMS, Human Monitoring System; HPMS, Human Performance Management System; HPSN, Human Performance Semantic Network; HSIM, Human Simulator; HCON, Human Controller; CRC, Control Room Controller; EEI, External Environmental Interface; PSF, Performance Shaping Factors.

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The IDCRC includes a significant number of subsystems that work together in an interoperable way to improve monitoring, decision support, and human-machine cooperation. The Human Performance Management System (HPMS) is at the heart of this architecture, and it is a decision-support system that constantly monitors, analyzes, and interprets human performance through physiological, behavioral, and environmental data. The HPMS estimates key performance measures, including stress level, workload, cognitive fatigue, and situational awareness, and relates them to operational parameters to assess human factors' influence on the overall performance of the plant.

Complementing this framework is the Human Performance Semantic Network (HPSN), which serves as the reasoning architecture behind the process of mapping Performance Shaping Factors (PSFs) to system structure, environmental context, and human behaviors; enabling the HPMS to make contextually informative inferences and identify performance trends based on the programmed relationships among operator state variables, procedural steps, and task environments. The influence and effect of performance shaping factors, situational awareness and workload on an operator's reliability has been extensively investigated and studied in frameworks of nuclear human reliability analysis (De Ambroggi and Trucco, 2011; Fernandes and Braarud, 2015; Gertman, 2005). Together, these systems will allow the IDCRC to move beyond the classical static design of Human-Machine Interface (HMI), toward an adaptive and context-aware interface whereby operator states and system behavior are dynamically connected.

The Human Monitoring System (HMS) serves as the primary sensor interface, obtaining physiological signals (such as heart rate variability, blink rate, eye gaze, voice tone, and posture) and environmental information (such as lighting, temperature, and noise levels) for the purpose of quantifying operator readiness and alertness levels. The Control Room Controller (CRC) coordinates operational feedback between the digital control system and the HPMS, while the Human Simulator (HSIM) allows for scenario-based evaluation of operator responses, team dynamics, and system load during simulated transient and normal operation. The External Environmental Interface (EEI) further extends this monitoring beyond the boundary of the control room to include external disturbances and environmental stressors to performance evaluation models.

Through the amalgamation of these units, the recommended IDCRC presents a closed-loop human performance feedback mechanism that needs ongoing assessment of operator state and system performance. It delivers once-lag alerts, adaptive display modifications, and decision-support actions to enhance situational awareness and lessen the possibility of human errors. To these ends, the proposed attributes accord with international human-factors and interface design foundations established in NRC human-factors review guidance, IAEA human-factors standards, and CSA nuclear design requirements (U.S. Nuclear Regulatory Commission, 2012; International Atomic Energy Agency, 2018; International Atomic Energy Agency, 2001; CSA Group, 2023a; International Atomic Energy Agency, 2013).

EEG, eye-tracking, and blink-based fatigue assessments have demonstrated the importance of physiological and behavioral monitoring for understanding operator workload and cognitive state (Joshi et al., 2023; Banholzer et al., 2021; Marquez, 2019; Steinhauser and Yeung, 2010; Hancock and Warm, 1989; Afzal et al., 2022; Braarud, 2024). Extending this body of work, the HPMS incorporates multimodal data, introduces machine learning methods and employs a unified analytical framework directly connected to the plant control logic so that performance may be estimated, and the system could respond in real-time. While the system can be used across reactor types, the present study is focused on CANDU-based SMRs, given a domestic fuel source, best available operational reliability with heavy-water moderation. The integrated HPMS-HPSN framework should ultimately improve safety, efficiency and resilience in SMR digital control rooms through a human-centered automation layer that incorporates human judgment and intelligence-enabled support.

1.1. Novel contributions

The current work on the IDCRC builds on the body of research related to nuclear plant control rooms and represents an evolution of the previous Digital Control Room (DCR) design by bridging together three main areas of research:

- 1) The development of a human/machine integrated environment in which a person can interact with a nuclear power plant in a collective manner.
- 2) The development of real-time, responsive, adaptive, and interactive control capabilities of a plant from the perspective of human components.
- 3) The development of a fully functional closed-loop system that combines the elements listed above into a unified operating system that allows the operator to continuously monitor, simulate and control the plant using either direct command or automated responses from the environment.

1. Multimodal Human-State Monitoring Integrated Directly into Plant Logic

DCR designs now include continuous, real-time monitoring of a plant's operating parameters and conditions; however, only a few DCR designs have integrated physiological (e.g., heart rate), behavioral (e.g., eye tracking, blink rate, posture, and voice stress), and environmental (e.g., temperature) human data to a plant's operational loop. The IDCRC model uses heart rate, eye tracking, blink rate, posture voice stress, and environmental parameters to determine the plant state (a direct correlation) through quantitative and continuous determination of human workload/stress/fatigue/situation awareness.

2. Semantic Reasoning Layer for Human-Performance Context Interpretation

Historical operator support systems used primarily rule and alarm-based logic to guide operators; by using a structured reasoning framework, the proposed HPSN maps PSFs against tasks and procedures, human state, and environmental conditions. This affords operators a contextually aware way of interpreting PSF data, predicting the likelihood of performance degradation due to PSFs, and detecting risk trends well in advance of observations or measures based purely on plant data.

3. Adaptive Interface Algorithms That Reconfigure the Control Room in Real-Time

As opposed to static or manually adjusted digital HMIs, the IDCRC implements a dynamic interface adaptation algorithm that modifies the layout and structure of the on-screen elements, the amount of information, prioritization of alarms, and visual complexity based on real-time operator state and cognitive load readings. This feature supports a dynamic workload regulation, that aims to minimize abrupt changes in attention during brief changes, helping operator focus and improves accuracy.

4. Integrating Environmental Controls into Human Performance Management

There has not been much research in existing DCR literature to date that addresses the area of workspace environmental control regarding human performance. The EEI introduced in this paper provides an intelligent adaptive control room environment that automatically adjusts the environmental controls (lighting, temperature, airflow, and acoustics) based on an operator's workload and alertness metrics.

5. Scenario Testing Assisted by a Closed Human-In-the-Loop Architecture

The research illustrates how multimodal monitoring; semantic reasoning, adaptive display logic, and environmental control work together to ensure stable performance by performing controlled shut-downs and turbine trip recovery scenarios. This provides an example of a complete integrated closed loop system rather than inspections, tests, or other types of isolated module tests.

6. Real-Time Assessment Enabled via the Integration with the SMR Simulator, iPWR

By integrating human factors monitoring and a real-time SMR simulator, the IDCR allows for monitoring operator physiological and behavioral responses during transient events, thereby allowing for an integrated evaluation framework that connects shifts in human performance with changes in plant dynamics. The combination of these two capabilities allows for the first time to clearly demonstrate the relationship between operator performance changes and plant dynamics in published SMR control room studies.

2. Integrated digital control room

Modern control rooms in Small Modular Reactors (SMRs) require adaptive systems to mitigate human error and enhance operational performance. This section outlines the architecture of a modular control system that integrates environmental sensors, cognitive adaptation, and

real-time procedural support designed to align with the evolving standards in Human Factors Engineering as identified in IAEA SSG-51, NUREG-0711, and CSA guidelines.

The IDCR (Integrated Digital Control Room) combines Human Performance Monitoring, Semantic interpretation, and adaptive interface control into one unified operational environment for SMRs. In contrast to standard Digital Control Rooms that consider human monitoring, interface behavior and plant dynamics as separate subsystems with limited inter-relationships, the IDCR provides an explicit closed loop architecture where human state, task demands and plant conditions are continuously interacting with each other, as shown in Fig. 1: Input-Output Signals between SMR and the Integrated Digital Control Room.

The framework is intended to accomplish two things. First, it helps ensure that operator physiological and cognitive conditions are in alignment with the demands of current plant operations. Second, it allows the interface and surrounding environment of the control room to automatically respond to any signs of operator overload, deteriorating situational awareness or increased stress levels. This framework is achieved through the operation of three primary components: human state estimation, semantic reasoning, and adaptive control.

The Human Monitoring System (HMS) receives input about an operator's physiological, behavioral, and environmental state from multiple modalities. These modalities are transformed by the Human Performance Management System (HPMS) and translated into quantitative indices of the operator's level of stress, workload and attentional

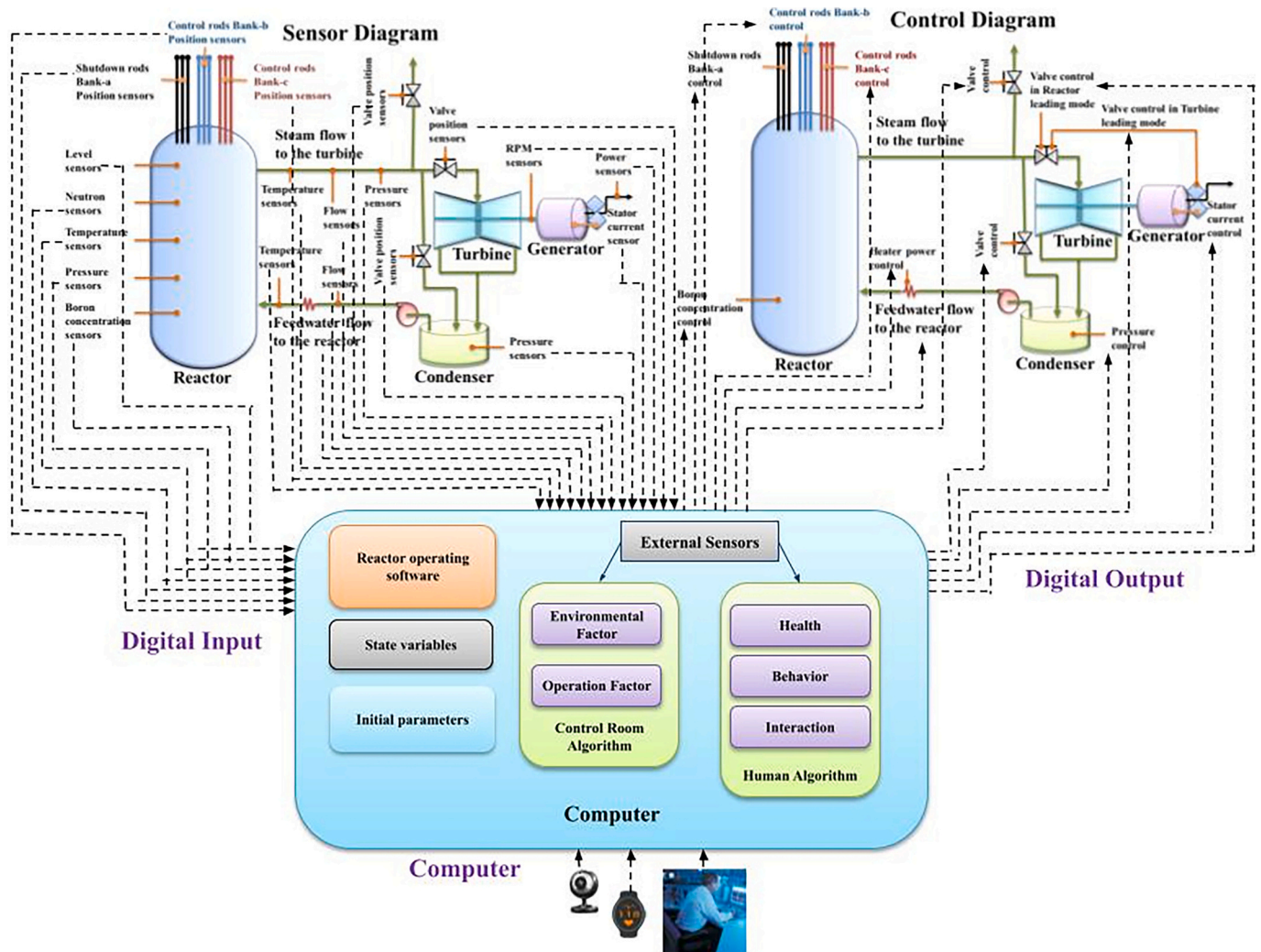


Fig. 1. Input-output signals between SMR and the integrated digital control room.

status. The Human Performance Semantic Network (HPSN) then analyses these indices in conjunction with task demands and the operator's known Performance Shaping Factors (PSFs) to determine how likely an operator is to experience verification errors, lapses in attention or deviations from procedures. The risk assessment generated by the HPSN is then transferred to the adaptive display & environmental control systems, which then change the level of complexity of the operator's interface, increase the visibility of critical indicators, and change the lighting / temperature levels in the control room to provide appropriate guidance to the operator so they can continue to perform at optimal levels.

The redesigned interface built for collecting and analyzing human performance data during simulator use. This can include eye tracking, decision speed, error rates, and interface navigation. These insights enable system improvements in display hierarchy, control positioning, and alert prioritization. The system also supports flexible layouts for single or multi-reactor operation, allowing custom control room design testing under realistic load conditions. Operator feedback loops are used to improve performance and reduce cognitive load. This mechanism allows the SMR controller to validate the actions of the operators in real-time and provides for continuous alignment between the human state and the behavior of the plant.

Additionally, these processes allow for a dynamic evolution of the IDCR during transient states, where a singular event that occurs at the plant changes the operator's state; the operator's state subsequently changes the interface behavior; and finally, the interface behavior continues to feed back the operator's performance.

3. Methodology

The Integrated Digital Control Room's (IDCR) method is based on an integrated digital control room closed-loop architecture that has real-time sensors and a machine learning-based approach for the operator's contextual awareness through semantic reasoning on what is happening; therefore, it provides a view of the operator's state, context, and environment as they operate in real time. The various methods of the Integrated Digital Control Room include Data Analysis Pipeline and Algorithms that allow the IDCR to be dynamic throughout the sequence of events occurring during Small Modular Reactor (SMR) operation. The IDCR is built on a five-part framework: (1) multimodal human-state estimation, (2) Semantic network-based semantic reasoning (Human Performance semantic network (HPSN)), (3) Adaptive display logic, (4) Data Fusion and synchronization pipelines, and (5) The Human-in-loop control loop.

3.1. Estimating human state

As part of the Human Performance Management System (HPMS), operator state estimation turns raw/physiological/behavioral signals created by the Human Monitoring System (HMS) into quantitated performance metrics. The HPMS framework normalizes deviations away from operator baselines to create stress/load indices based on the average or median of operator deviations from those baselines

1. Physiological
2. Behavioral
3. Environmental
4. Situational

From the raw data collected by the HMS, the processing of raw data is represented as a set of functions of time including but not limited to:

- $HR(t)$ = heart rate
- $B(t)$ = blinking frequency
- $F(t)$ = fixation of gaze
- $P(t)$ = posture variation

Raw data is processed through noise and distortion removal, divided into filters of 5 s. Each data signal variable is then normalized relative to a baseline retrieved from the HPSN.

$$X_{norm}(t) = \frac{X(t) - X_{baseline}}{X_{baseline}}$$

The stress-index formulation is derived from neuroergonomics principles proposed by Zhang, Wang, Chen, and Li (Zhang et al., 2024):

$$SI(t) = 0.35HR_{norm}(t) + 0.25B_{norm}(t) + 0.20P_{norm}(t) + 0.20E(t),$$

It uses data collected through $E(t)$, which is an indication of the amount of environmental discomfort experienced by a user. It includes a variety of environmental variables (e.g., temperature, light) that differ from acceptable levels for user performance. The load index combines both attentional dispersion and the stress index:

$$WI(t) = 0.5[1 - F(t)] + 0.5SI(t).$$

This formulation allows for real-time assessment of both cognitive load and psychophysiological stress, thereby forming the basis for semantic reasoning and the reactor interface adaptation. A simplified pseudocode representation is included for reproducibility:

```
function HPMS_Estimate_State(human, env):
    HR_norm = normalize(human.HR)
    Blink_norm = normalize(human.blink)
    Posture_norm = normalize(human.posture)
    Fixation_score = 1 - human.fixation_dispersion

    SI = 0.35*HR_norm + 0.25*Blink_norm + 0.20*Posture_norm + 0.20*env.
    stress_factor
    WI = 0.5*Fixation_score + 0.5*SI
    return SI, WI
```

3.2. Semantic reasoning using the human performance semantic network

The HPSN allows for organized representation of how Performance-Shaping Factors (PSFs) influence the operator task context and operator state/environmental conditions. The network can be created as a Weighted Directed Graph, with nodes representing variables (stress, workload, verification demand, time pressure, and task criticality), and edges denoting empirically derived causal relations identified in NUREG-0711, SPAR-H and other Human Factors research publications.

At each point in time, the mapping of the state vector into the PSF cluster closest to it will be determined, and the likelihood of a performance decrement evaluated.

$$S(t) = \{SI(t), WI(t), F(t), TaskContext(t)\}.$$

A scoring function, $R(t)$, based on the weights assigned to active PSFs will be calculated.

$$R(t) = \sum_i w_i PSF_i(t),$$

where w_i reflects the relative contribution of each PSF to error-likelihood in the current task context.

Semantic Reasoning helps to identify when the Operator may be entering into a high risk (e.g., Lowered Verification Accuracy or Increased Omission Probability) state. At the same time, a further simplified reasoning routine is included below.

```
function HPSN_Reason(SI, WI, task):
    PSFs = activate_nodes(SI, WI, task)
    Risk = weighted_sum(PSFs)
    if Risk > 0.7: return "HighRisk", Risk
    if Risk > 0.4: return "MediumRisk", Risk
    return "LowRisk", Risk
```

3.3. Adaptive interface logic

The adaptive display system alters the structure and the visual density of the control room interface based upon the operator's state, as well as the alarm conditions present. The ADS has two classes of triggers that cause it to perform these actions:

1) Intrinsic triggers are based on Human State Indices (such as elevated stress and/or workload levels).

2) Extrinsic triggers include alarms or transients generated by the operations of the plants.

As the workload increases, the adaptive display system will increase operator usability by reducing perceived cognitive load by collapsing secondary panels into one or more areas, increasing contrast on critical indicators, consolidating information on verification-required items, and enlarging key procedural steps. Under stable/low workload conditions (i.e., low workload), the ADS will revert back to the default mode. The core adaptation policy is expressed as:

$$Layout(t+1) = \begin{cases} L_{alarm}, & \text{if alarms active} \\ L_{simplified}, & \text{if } SI(t) > 0.5 \text{ or } WI(t) > 0.5 \\ L_{default}, & \text{otherwise.} \end{cases}$$

An algorithm is included for clarity:

```
function ADS_Update(SI, WI, fixation, alarms):
    if alarms.active:
        display.activate_alarm_layout()
    return
    if SI > 0.5 or WI > 0.5 or fixation < 0.4:
        display.simplify()
    else:
        display.restore_default()
```

This logic reflects the behavior observed in the turbine-trip scenario where the system automatically merged several status indicators into a single summary panel when workload increased. Adaptive and decision-support systems have been shown in prior research to be beneficial in nuclear control rooms for enhancing operator performance, handling of alarms, and recovery under high cognitive demands. (Alberti, 2023; Jeong, 2024; Kim et al., 2014; Mietkiewicz, 2024).

3.4. Data-fusion and synchronization pipeline

Integrating all human, environmental and plant data streams creates a unified fusion pipeline where plant state data produced by the SMR simulator will be sampled at 1 Hz, while the HMS will produce the bio signal feature data at faster rates (i.e. higher than 1 Hz). All of this data is resampled to align the timestamps of all the collected data before it is sent to the Reasoning Engine and Adaptation Engine. The synchronized state at time t is

$$X(t) = \{Plant(t), Human(t), Env(t), SI(t), WI(t), Risk(t)\}.$$

The fusion loop proceeds as follows:

1. Acquire plant, human, and environmental measurements.
2. Compute stress and workload indices.
3. Assess risk through the HPSN.
4. Update interface and environmental setpoints.
5. Log all variables for after-action analysis.

The full cycle, executed once per second, is represented by:

```
function IDCR_Fusion_Loop():
    plant = SIM.read()
    human = HMS.read()
    env = EEL.read()

    SI, WI = HPMS_Estimate_State(human, env)
    RiskState, RiskScore = HPSN_Reason(SI, WI, plant.task)
```

(continued on next column)

(continued)

```
ADS_Update(SI, WI, human.fixation, plant.alarms)
EEL_Adjust(SI, env)
Logger.store(plant, human, env, SI, WI, RiskScore)
```

3.5. Human-in-the-loop operational cycle

IDCR provides operators with real-time intelligent assistance (a method that enables a person to continue performing their job duties). The operational cycle begins with the operator interacting with a simulator to replicate actual plant dynamics. As a result of these interactions, operators react with both physiological and behavioral responses, which are captured by the HMS. The HPMS and HPSN interpret these responses and modify both the way operators visually interact with the plant and the way they feel while interacting with it. The resulting adaptation will affect how operators behave in future actions, completing the operational cycle.

Unlike traditional supervisory control systems that simply estimate human performance as an arrangement of tasks, IDCR finds a way to include the estimation of human performance directly into the logic for synchronising plant operators. Because of this, IDCR allows for adaptive human-machine coordination instead of only providing a mechanism for static or alarm-based operator interactions.

4. DCR main modules

The proposed Integrated Digital Control Room (IDCR) design encompasses main modules as shown in Fig. 2.

This outline of the IDCR includes the twelve interoperable modules providing a collective system of plant simulation, operator state monitoring, humans interacting with their environment through adaptive interfaces, and environmental control systems developed as a closed-loop system. The IDCR's respective components work together to facilitate continuous plant, operator, and environmental monitoring and interpretation of responses, thus allowing the control room to operate as a "smart" human-machine environment versus a static interface.

Of the research conducted, the closed-loop approach is a significant finding. While the architecture combines operator states, operational contexts, and plant dynamic states, it provides the basis for the adaptive operations assessed in the case study. Adaptive operation capabilities assessed during the case study included interaction with a workload-aware interface that adapts based on operator workflow demands, performance-informed decision support capabilities when troubleshooting, and environment-mediated deferrals of corrective action. Findings of this research highlight that a unified architecture provides significant improvements to operator interactivity and resilience of advanced reactor control rooms.

4.1. DCR functions

The Integrated Digital Control Room (IDCR) encompasses the fulfilment of eight distinct functional capabilities: display, interface, guide, validate, monitor, diagnose, execute, adapt, train and report through a time-based operation that enables real-time coordination of plant dynamics and operator states with the growth of interface behaviors that evolve in response to how the operator interacts with the IDs. This evaluation does not focus on the mechanics of subsystems; instead, it focuses on the operational capabilities that will enable Small Modular Reactors (SMRs) to support situational awareness, workload stability, error mitigation, and recovery performance. Details of the functional mappings and parameter definitions are included in Appendix A.

The mapping between main functions and modules is illustrated in

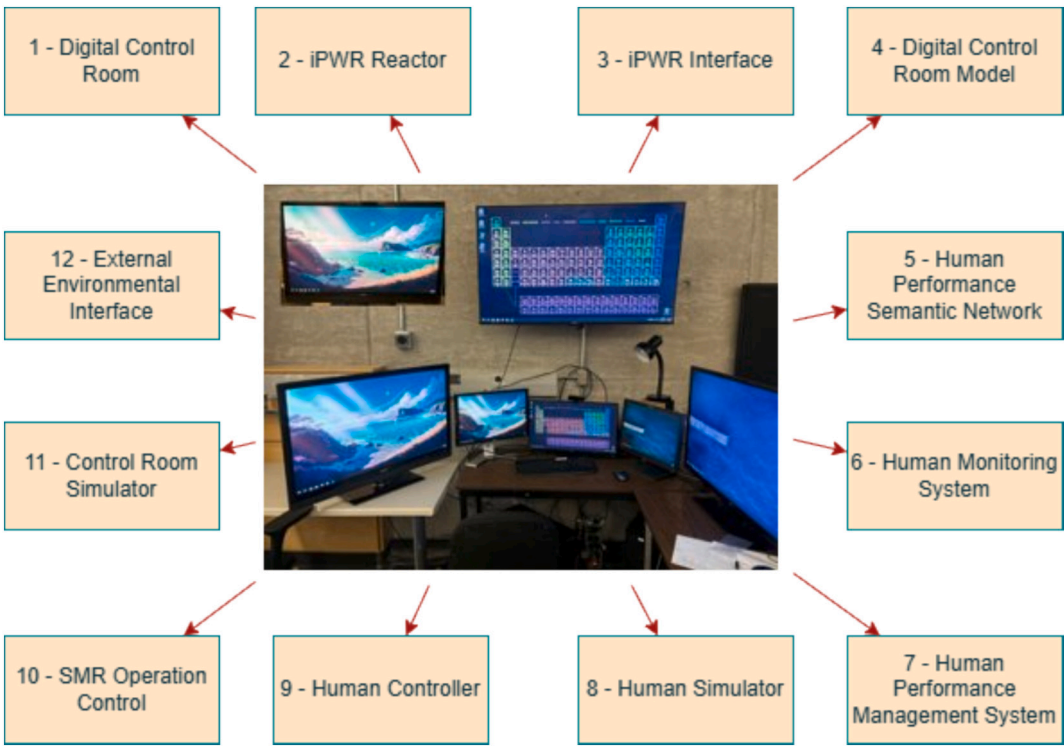


Fig. 2. The proposed integrated digital control room design (DCR).

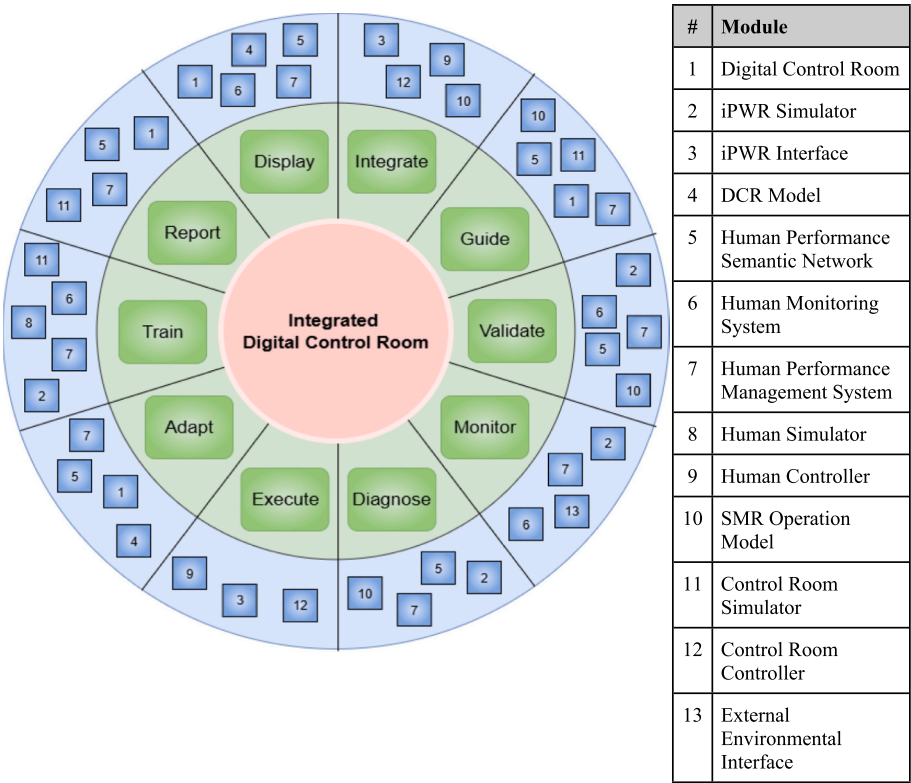


Fig. 3. Integrated DCR functions coupled with main modules.

Fig. 3: Integrated DCR Functions Coupled with Main Modules.

#	Module
1	Digital Control Room
2	iPWR Simulator

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3	iPWR Interface
4	DCR Model
5	Human Performance Semantic Network

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6	Human Monitoring System
7	Human Performance Management System
8	Human Simulator
9	Human Controller
10	SMR Operation Model
11	Control Room Simulator
12	Control Room Controller
13	External Environmental Interface

The IDCRC defines the relationship between plant data, human performance information/procedural instructions and adaptive controls in a closed-loop architecture by establishing the operational categories of display, integrate, guide, validate, monitor, diagnose, implement, adapt, train, and report. The operational categories help describe how the system provides situational awareness and aligns human and plant activity while providing adaptive decision support. Appendix B provides a technical mapping of the operational categories to system modules; however, the main purpose of the functions is to help illustrate the ability of the IDCRC to provide a coordinated interface between humans and machines during operations involving an SMR.

Together, Fig. 2 and Fig. 3 provide architectural grounding for the closed-loop methodology described in Section 3, while the detailed modules and layout diagrams are included within Appendices E–H.

4.2. Design principles

This research on closed-loop behaviors is based on human-factors and control-room design principles that promote ease of use, modularity, and flexible user input. The suggested control room design complies with national and international guidelines for SMR design and human factors engineering, such as CSA human-factors standards and IAEA technical reports (CSA Group, 2023b; International Atomic Energy Agency, 2016). These principles are the foundation for creating an ergonomic and architectural environment that is designed to facilitate the use of closed-loop behavior, while detailed descriptions of how these principles were applied to the IDCRC are provided in Appendix C.

4.3. System design

Three main subsystems facilitate the adaptive Human-Machine Interface (HMI) and real-time feedback to the operator via the IDCRC. The main subsystems of the IDCRC are: 1) the Human Controller (HCON), 2) the Control Room Controller (CRC), and 3) the External Environment Interface (EEI). The subsystems convert the operator's commands, the system response to those commands, and workspace conditions into structured information that is used by the organism-state estimation, semantic reasoning, and adaptive-display methodologies previously discussed.

4.3.1. Human controller

The HCON establishes connections between the operator's physical input and their physical and mental status so that the IDCRC may better understand operator actions based on workload, stress, and situational awareness. Essentially, while the HCON could be used as a controller per se, it is contextual and recognizes when operators are experiencing lapses in the verification process, delays in responding to commands, or deviations from their normal anticipated patterns of interaction, which often suggests a heightened cognitive load on that operator. The complete description of parameters, evaluation metrics, and module functionality is listed in Appendix D.

4.3.2. Control room controller

The primary function of a CRC is to act as the main engine where all execution and validation of commands happen inside the IDCRC Structure. This means that the controller synchronizes user input with

commands sent from the system as well as the states of the simulator. By following this process, all interactions with the simulator will follow the rules of the nuclear facility's safety conditions and general logic. With the CRC, the operator's command will be verified on a real-time basis; therefore, ensuring the operator's command is routed safely into the simulator. Appendix E provides information on the internal processes, communication metrics, and functional schema of the CRC.

4.3.3. External environmental interface

The EEI ensures that environmental factors such as temperature, humidity, noise level, etc. can be monitored and maintained under optimal conditions to keep the operator comfortable and maintain cognitive readiness. By using these environmental measurements and relating them to human-state indicators, the interface helps maintain operators' alertness levels and reduce the effects of physiological strain during high-demand scenarios; this helps preserve operator performance during critical situations. Information on the detailed environmental parameters, performance thresholds, and necessary adaptation rules are outlined in Appendix F.

In combination, these subsystems provide the necessary operational framework for the IDCRC to support adaptive behavior, sustain human performance stability over time and across multiple uses, and facilitate plant-human interactions throughout the entire transient and nominal operations of the SMR.

4.4. Design layout & interfaces

The Multi-Display Station and Wall-Screens for an Integrated Digital Control Room (IDCRC) provide a way to adaptively reconfigure the mechanical systems, human performance, and process-and the associated data. The Multi-Display Station (MDS) and Wall-Screens do not have fixed physical locations, rather they dynamically redistribute the computer screens for maximum visibility and the least amount of cognitive pressure on the operator while maximizing operator situational awareness during process/transient conditions, based upon human state readings (i.e., workload, fixation, stress) in addition to plant conditions of operation as the system operates. The Human State readings and Plant conditions are directly integrated into the Mapping of the displays, forms, and sensor-actuator integrations for the MDS and Wall-Screens. Please see Appendix G for complete mapping information on the displays, forms, and sensor actuator integrations for the IDCRC.

4.4.1. Performance measurement framework

The performance of the IDCRC was evaluated using the human, system, and interaction measures according to human-factors guidelines and adaptive interface research. These measures provided insight into operator response time, error frequency, situation awareness, workload and training, and through these measures the closed-loop nature of the IDCRC's human-machine interaction can be evaluated in both steady and varying modes of operation (nominal vs. transient). The metrics demonstrate the effectiveness of the IDCRC in maintaining a rigidly defined human-machine interaction despite fluctuations in plant loads and changes in operator state. The study's workload and task demand evaluation approaches are consistent with accepted techniques for workload classification and cognitive task analysis (Kniesly et al., 2021). A complete listing of performance measures and parameters that support them can be found in Appendix I.

5. Scenario execution using SMR simulator

a. SMR Simulator overview

The Small Modular Reactor (SMR) Simulator is the core computational engine of the system, modeling the physics, system dynamics, and response algorithms of a pressurized water reactor. It processes operator commands, tracks parameters, and generates real-time responses for

both manual and automated control. This foundation supports a wide range of scenarios, from normal operations to abnormal events. Within the control room, the simulator is central for training and preparation. It replicates the behavior of SMRs under various conditions, provides instant feedback, triggers alarms, and predicts responses. By simulating thermal-hydraulic processes, control logic, and component interactions, it gives operators practical experience with plant dynamics, safety procedures, and fault detection, helping to reduce human error and build confidence in managing severe situations. The simulator also functions as a verification and design tool. It allows real-time evaluation of interface changes, alarm prioritization, and adaptive controls, showing how operators and systems respond together. It has been demonstrated that simulation-based training and assessment greatly enhances human performance, safety, and procedural compliance in high-risk operational environments (Bienstock and Heuer, 2022). This makes it invaluable for engineers and designers aiming to test new digital control room features and decision-support systems. Ultimately, it bridges theory and application, ensuring fault-tolerant, ergonomic, and safety-compliant designs in line with CNSC REGDOC-2.5.1 and IAEA SSG-51.

b. Interface system

The User Interface and the Interface System define how users interact with the basic simulator. The User Interface is a standalone piece such as a control panel or monitor display that provides a particular operator with access to helpful information, allows data entry, and accepts results. In contrast, the Interface System refers to a broader, networked system of these user interfaces. This system enables multiple operators or stations to work in conjunction, share information, and maintain synchronized control of different parts of a simulator, ensuring seamless interaction within a shared simulation environment. The simulator was integrated with the digital control room to support real-time plant emulation, human performance monitoring, and adaptive interface testing. Work also included linking the simulator with the operator Human Monitoring System (HMS) and the Human Performance Management System (HPMS) with the Human Performance Semantic Network (HPSN) reasoning module, so that plant events could be analyzed alongside biometric and behavioral data. Closed-loop test scenarios, such as shutdowns, startups, and turbine trips, were executed to evaluate both plant responses and human-machine interactions. The simulator interface design details are presented in Table 1.

The proposed interface system structure is explained in Table 2, which details the primary components of the interface system and their corresponding properties and parameters.

Table 2
Interface system structure.

Component	Key Elements
Human Monitoring	- Eye tracking (fixation, blinking, gaze direction) – Facial emotion recognition - Heart rate and breathing rate - EEG for detecting cognitive workload and emotional state
Human Performance Assessment	- Behavioral models (e.g., stress, fatigue, distraction) - Algorithms to analyze physiological inputs and translate them into performance states
Feedback and Adaptation Module	- Interface adjustments (visual complexity, alarm loudness, task support) – Actionable responses (task redistribution, supervisory alerting) - Safety margin estimation and risk mitigation procedures
Control Room Integration System	- Links adaptive feedback to digital control room systems - Interfaces with SMR simulators and emergency control stations
Data Logging and Analytics	- Human Performance Semantic Network database storing long-term biometric and task-related data - Enables continuous improvement and training customization

6. Case study

The proposed case study is related to detection of malfunction during Small Modular Reactor (SMR) operation. The main analysis of main subsystems, functions, and related diagnosis is shown in Table 3. (See Table 2.)

This feature interfaces seamlessly with the Control Room Controller (CRC) and Human Performance Management System (HPMS) to record every detected failure and the human response metrics of reaction time, vocal confirmation, and eye fixation. It adds realism to the simulator with an AI-assisted diagnostic feedback loop that supports faster detection of complex multi-signal failures and provides systematic and traceable data for performance assessment and training analysis.

The test process begins with continuous monitoring of simulator data and alarm streams. Once any alarm or visual indicator changes, the system aggregates the current set of active alarms and checks them against the malfunction rules stored in the database. When a rule pattern matches, the malfunction detection engine identifies the specific fault, notifies the operator via an on-screen popup, and records the event details in the HPMS/HPSN logs for later analysis. This **closed-loop diagnostic cycle** ensures that both technical and human responses are captured for performance evaluation and training improvement.

The proposed test flow chart is shown in Fig. 4.

The malfunction detection testing is performed based on the proposed algorithm for the adaptive user interface, as shown in Fig. 5. The image shows a malfunctions alert system engaged in a Steam Header Break failure. The system effectively monitors alarms and plant process indications displayed in the simulator interface. If an active alarm set (e.

Table 1
Simulator interface system design details.

● Design & Functions: ○ Session control (initial conditions, checkpoints, timeline management). ○ Scenario definition (parameter sets for normal, abnormal, and faulted operations). ○ Real-time execution of operator commands (setpoints, valve actuation, pump starts). ○ Data streaming of process variables (temperature, pressure, flow, neutron flux). ○ Alarm/event broadcasting for deviations and safety trips. ● Mechanisms: ○ Continuous closed-loop interaction between plant states, operator inputs, and adaptive interface adjustments. ○ Algorithms to evaluate interlocks, safety margins, and recovery sequences under varying conditions. ○ Optical and data-tag verification of interface states to ensure displays are accurate. ● Integration in Control Room: ○ Linked with adaptive HMI for feedback and alerts. ○ Coupled with HPMS/HPSN for human performance analysis. ○ Supports training simulators for structured learning and evaluation. ● Closed-Loop Scenarios: ○ Automatic trips and recovery sequences tested by injecting faults. ○ Operator responses monitored in tandem with plant state progression. ○ Performance shaping factors correlated with system dynamics to assess resilience.
--

Table 3

Functional components of the automated malfunction detection system.

Subsystem	Function	Example Trigger Inputs	Automated Diagnosis	Interface Response
Alarm Parser	Continuously scans and parses active alarms from simulator UI	Reactor Trip, Gen Breaker Open, ADS Actuation	Cross-matched to <i>Inadvertent ADS Initiation</i>	Popup alert with malfunction title and affected systems
Rule Engine	Applies pattern-matching logic using JSON rule base	Reactor Trip, Turbine Trip, Low RCS Pressure	Identified as <i>Steam Header Break</i>	Displays detected malfunction banner and logs event
UI Monitor	Analyzes color-coded display elements (e.g., red/emergency indicators)	ADS - > Emergency, FWS - > Isolated	Matches high-priority failures	Triggers visual emphasis on affected panels
Notification Manager	Sends interpreted result to operator and system logs	Matched malfunction label	Fault classification and time stamp	On-screen popup and HPMS/HPSN data entry

g., Reactor Trip, Turbine Trip, MSS Isolation, FWS Isolation, PDHR Actuation, Low RCS Pressure, etc.) matches the predefined rule pattern in the malfunctions rule base, the interface automatically generates a faulty equipment notification pop-up that notifies the operator of a malfunction. The notification will list the active alarms that resulted in the match and will help identify the incident in the appropriate, traceable context for verification purposes. The notification will automatically log with timestamps and reference the event to HPMS/HPSN datasets for evaluating operator response time, situational awareness, and confirmation accuracy.

Fig. 6 presents the three-tier integrated diagram linking human parameters, control room environmental conditions, and SMR process events. The upper tier shows variations in heart rate, skin temperature, and HPMS workload index; the middle tier tracks room temperature, humidity, and lighting; and the lower tier marks simulator steps. The diagram demonstrates that changes in operator workload align with process events rather than environmental shifts. This confirms that the proposed Integrated Digital Control Room (IDCR) design effectively separates cognitive workload factors from environmental influences while maintaining stable physical conditions. Unlike conventional CANDU main control rooms, which rely on isolated analog indicators and post-event analysis, the IDCR architecture enables real-time coupling between human, environmental, and process data, supporting adaptive displays and proactive workload management.

Fig. 7 shows the human-control room heatmap, which compresses six continuous parameters into a temporal-spatial view. Warmer color zones in workload coincide with lower attention levels mid-scenario, indicating increased cognitive demand. The lower environmental rows remain constant, confirming that workload changes originated from operational tasks rather than environmental factors. This fused visualization allows quick identification of correlations between human and system conditions, offering a capability not present in traditional control rooms. It supports real-time monitoring and human-performance-based decision-making rather than relying on static numerical data.

Together, these visualizations demonstrate how the proposed IDCR supports integrated situational awareness, workload management, and adaptive human-system interaction beyond what is achievable in legacy analog or semi-digital control room environments.

An extensive evaluation of the subsystem level was conducted on the Human Controller (HCON), Control Room Controller (CRC), External

Table 2

Integrated summary: scenario comparison and operator response.

Aspect	Scenario 1 – Controlled Shutdown	Scenario 2 – Turbine Trip and Recovery	Design Value / Takeaway
Overall System Behavior	Linear, predictable sequence; minor workload fluctuations during boron injection; steady physiological readings.	High cognitive load, multiple alarm bursts, and manual interventions created elevated stress and slower validation.	The IDCR effectively stabilizes human performance in both low- and high-load operations through adaptive interface and monitoring feedback.
Human-System Interaction	Grouped data displays and logical sequence helped operators focus and reduce visual scanning.	Alarm floods caused temporary confusion; adaptive interface reprioritized alarms and hid non-critical fields to restore clarity.	The modular interface enables quick reorganization of data forms to reduce cognitive overload and maintain control awareness.
Operator Workload and Stress (HMS / HPMS Data)	Heart rate and blink rates remained near baseline (WI $\approx$ 0.08).	HR +20%, blink +60%, stress index $\approx$ 0.7 during trip; recovered to $\approx$ 0.3 after adaptive cues.	The integrated HMS-HPMS loop detects physiological strain early and deploys interface and environmental adjustments to mitigate it.
Situational Awareness (SA)	SA $\approx$ 0.87, maintained throughout steps.	SA dropped to 0.68 during alarm floods, recovered to 0.80 after adaptive simplification.	Real-time SA tracking allows interface and simulator modules to regulate operator focus and maintain procedural compliance.
Reaction Time and Accuracy	Average command response $\approx$ 2.1 s; 100% accuracy in task sequence.	Average command response $\approx$ 3.0 s; 92–95% accuracy under stress.	The IDCR's procedural guidance and visual prioritization minimize delay even in high-pressure phases.
Adaptive System Responses	Minimal adaptation triggered, system remained in baseline configuration.	Adaptive Display compressed non-critical fields; lighting reduced 10%; simulator slowed to 0.9 $\times$ for cognitive pacing.	Adaptive controls demonstrate effective real-time compensation for stress and fatigue indicators.
Environmental Conditions	Room temperature 22 °C, 520 lx, comfort index 0.94.	Lighting automatically dimmed to 470 lx, airflow increased by 5%.	External environmental interface supports physiological comfort and alertness under varying cognitive loads.
System Integration (HPSN / Simulator)	HPSN linked Performance Shaping Factors (PSFs) to low-risk state (composite 0.92).	HPSN mapped elevated time pressure $\rightarrow$ error-likelihood link (composite 0.85).	Semantic reasoning enables early detection of risk nodes and causal prediction of operator behavior trends.

(continued on next page)

Table 2 (continued)

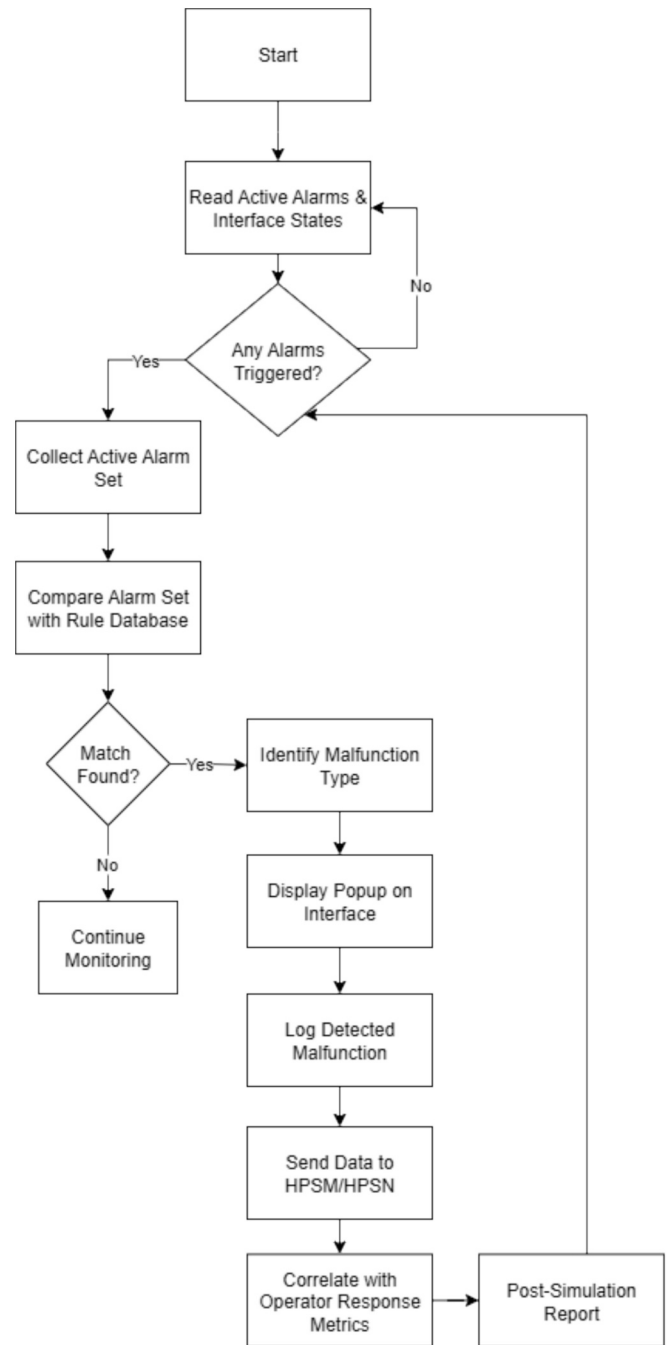
Aspect	Scenario 1 – Controlled Shutdown	Scenario 2 – Turbine Trip and Recovery	Design Value / Takeaway
Training and Procedural Reinforcement	Procedure execution aligned with baseline; used for model calibration.	Deviations logged and analyzed for training optimization and feedback via Human Simulator (HSIM).	Real scenarios reinforce adaptive training modules for procedural improvement.
Design Takeaway Summary	Stable human-machine performance at nominal workload.	Demonstrated resilience under elevated stress.	The IDCR design achieves operational stability through integrated human monitoring, adaptive interface logic, and environmental feedback.

Environmental Interface (EEI), iPWR simulator interface, and Adaptive Display System. From these evaluations, it was possible to see how the amount of workload for an operator, their interaction patterns with an interface, and the environmental conditions around them caused adaptive behaviors to occur within the IDCR. For example, high stress levels and a lack of fixation stability triggered a consistent interface simplification, and low levels of illumination caused environmental adaptations to be put into place to allow for greater visual comfort. The communication metric information generated by the CRC did not vary according to the testing scenarios, and this attested to the reliability in the routing of signals and the validation of commands. Appendix H contains all the detailed quantitative metrics, observations of performance, and sub-system response methods.

The case study included two test scenarios where the comparison showed key benefits of the proposed integrated control room with improved operation prediction and resilient performance under stress, as shown in Table 4: Integrated Summary: Scenario Comparison and Operator Response. Scenario 1 confirmed the IDCR's efficiency for standard, low-stress operations. Scenario 2 illustrated the strength of the design, namely, dynamic adaptations to human state and scenario complexity. Collectively, the results suggest that the IDCR's core advantage is the combination of human monitoring, adaptive display logic, and environmental feedback that promotes safe and efficient control under elevated cognitive and procedural stress. These results show that the IDCR's core advantage lies in adaptive combination. The human-system loop consisting of the Human Monitoring System (HMS), Human Performance Management System (HPMS), Human Performance Semantic Network (HPSN), Simulator, and Adaptive Display actively detects workload, simplifies data presentation, and stabilizes operator focus in relation to plant dynamic performance. The design supports safe, efficient control in steady and transient operating states, providing proof of the system's ability to combine plant dynamics, human factors, and adaptive response under the umbrella of digital integration in the control room.

6.1. Limitations of case study and experimental validation

Though the case study establishes the operational feasibility of the IDCR, there are many limitations to the current validation that must be noted. One such limitation was that the human-state measurements used to develop the Adaptive Interface were produced using simulation of physiological/behavioral data, rather than collection from actual operators. The methodology employed provided a controlled environment for testing/development as well as deterministic behavior but did not


Fig. 4. Malfunction detection logic flow.

provide an accurate representation of the variability, noise characteristics, or human adaptation strategy observed when operators perform tasks in the field. Therefore, the Workload and Stress Indicators presented in this paper reflect modeled tendencies rather than empirical psychophysiological response data from operators.

Two scenarios of the evaluation are represented as follows: Controlled Shutdown, Turbine Trip Recovery discussed closed-loop interactions that occur between human state estimation, semantic reasoning and adaptive interface behavior. These two scenarios were selected to demonstrate closed-loop interactions. However, neither scenario encompasses the entire array of transient or extreme conditions that are critical to SMR Operations. Other potential failure modes, such as Feed Water Loss, Small Break LOCA, Steam Generator Tube Deterioration and Complex Multi-alarm cascades are equally as important for

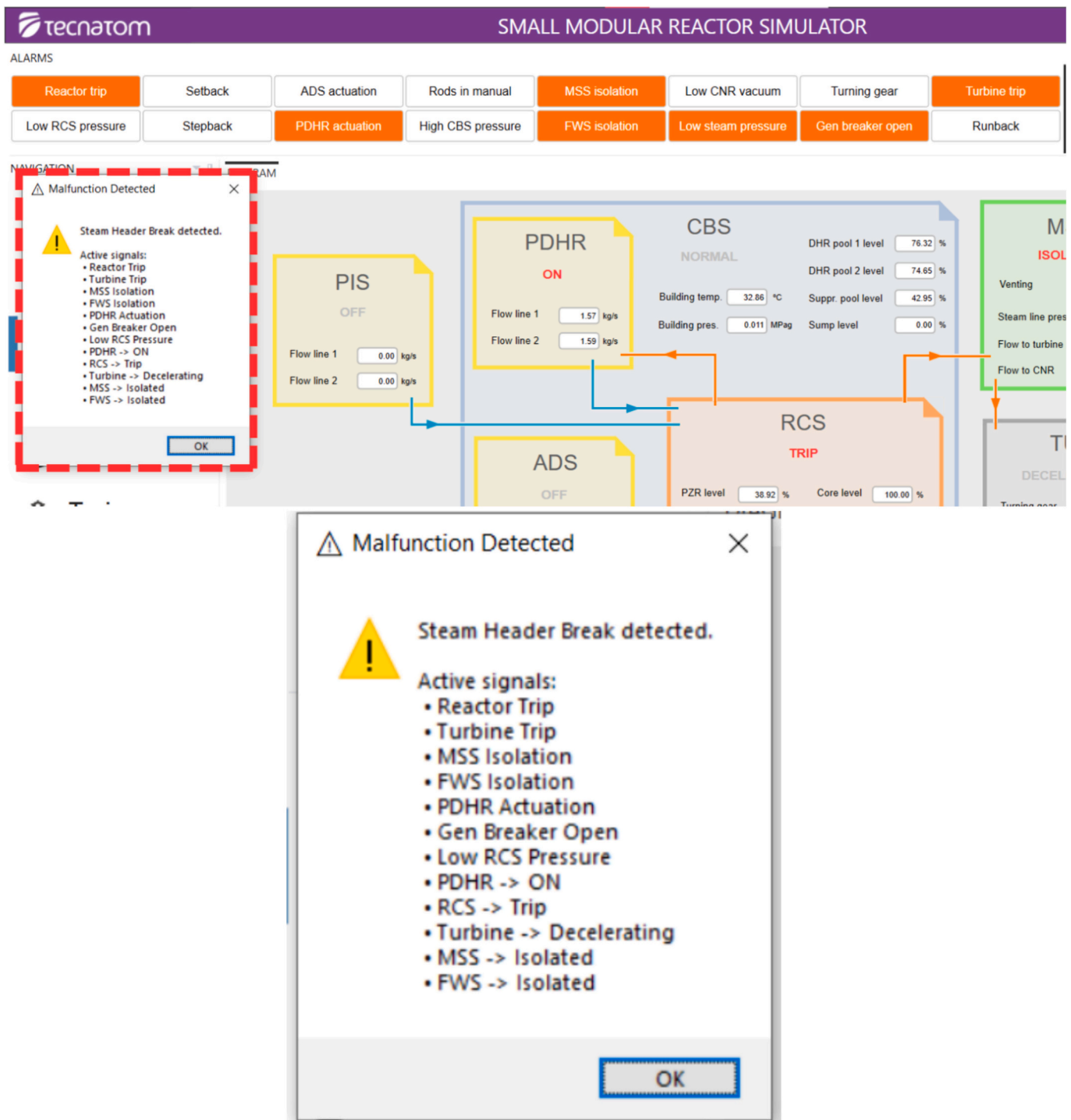


Fig. 5. Real-time malfunction detection popup in the SMR simulator interface.

evaluating the robustness of Adaptive Mechanisms under higher levels of cognitive and procedural requirements. As such, questions remain unanswered regarding the robustness of Adaptive Mechanism capabilities under high-level cognitive and procedural requirements.

Future work will incorporate a larger scope for validation, including both (1) a wider variety of high-consequence and multi-fault scenarios that test the limit of the predictive reasoning and adaptation algorithms; and, (2) controlled human-subject experimentation where actual physiological signals (e.g., heart rate variability, gaze dynamics) and behavioral reactions (e.g., responses to transitory events) are collected from human subjects. Using real operator data will allow the HPMS and

HPSN frameworks to be recalibrated against actual operator performance and increase the ecological validity of the frameworks. These results will help provide a more rigorous empirical basis for supporting the benefits of adaptive digital control rooms in SMRs.

7. Conclusion

An Integrated Digital Control Room (IDCR) concept has been created to combine various forms of monitoring and controlling the human working condition in the operation of Small Modular Reactors (SMRs) through a 'closed-loop' operation method. This approach has been

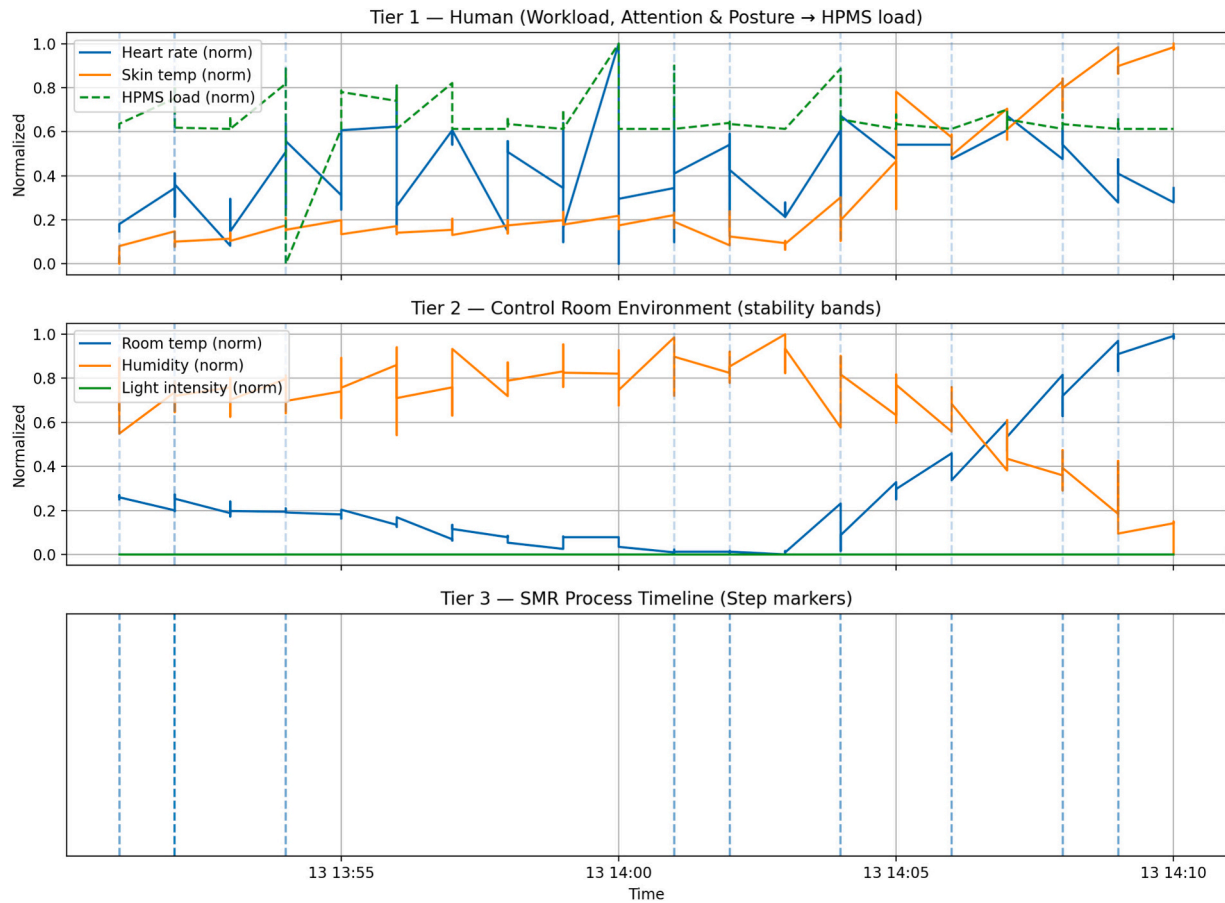


Fig. 6. Three-tier integrated diagram: human, control room, and SMR process.

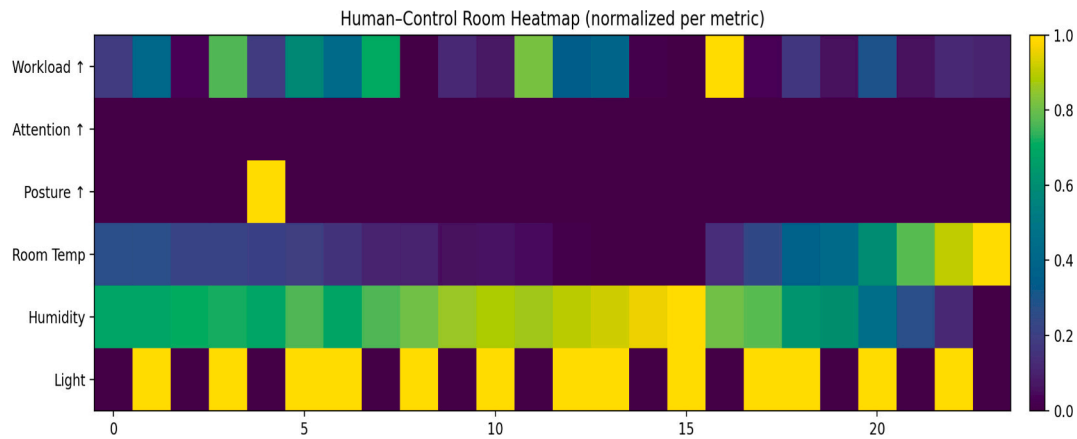


Fig. 7. Human control room heatmap.

validated through two types of test cases (methodologies/scenarios), where the combination of the Human Performance Management System (HPMS) with the Human Monitoring System (HMS)/Human Performance Semantic Network (HPSN) and adaptive logical interfaces can help stabilize operator performance under conditions of both low demand and higher levels of stress. When utilising these adaptive approaches in the two test cases, there were improvements in operator workload management, recovering situational awareness, and verifying tasks' accuracy. This supports the premise that human-centric digital control rooms serve to enhance efficiency and resilient operations. Despite the positive developments, many limitations exist that are currently placing limits on how the current architectures have been

deployed. The first limitation is the fact that the human-state data utilized in this study was developed using simulated input rather than through actual operators; thus, the results obtained cannot be considered ecologically valid as they pertain to human physiological and behavioral responses.

Second, when the architecture was originally designed, it was done so under a single-operator configuration. As such, the actual SMR control rooms consist of teams of multi-operators communicating with each other, having shared mental models, and coordinating their activities. It is unknown how these factors may positively or negatively impact adaptive interfaces in the real-world and could introduce some nonlinearities.

Third, the architecture has thus far not been subjected to stress testing with strict real-time constraints; therefore, a thorough evaluation of all the different latency budgets and computational reliability for a large scale full digital plant-grade integrated system needs to be conducted to fully understand the challenges of scaling these architectures.

Finally, long-term adaptability; how the architecture adapts to a changing operator proficiency as time passes; has yet to be investigated.

To address these challenges in future research, controlled human subject experiments will be utilized to validate the physiological monitoring and adaptation triggers as well as to extend the architecture so it can support multiple operators working together, and will be used to benchmark closed-loop algorithm real-time performance at high simulator loads. Also being explored will be using reinforcement learning methods to provide longer duration personalized interfaces, along with developing predictive modeling for early anomaly detection in human and/or plant states. These steps will provide additional scientific rigor to support deployment of adaptive, human-centered control rooms designed for next-generation SMRs.

This work establishes the potential for the building of integrated closed-loop digital control rooms using simulation, while further steps will be taken to evaluate the effects of building this type of room on real human subjects, conduct additional testing using complex and multiple failures, and to develop this type of room for application with multiple operators under real time.

CRedit authorship contribution statement

Hossam A. Gabbar: Writing – review & editing, Validation,

Appendix A. Appendices

A.1. Appendix A – DCR Main Functions

Table A1
DCR Main Functions.

● Display plant, human, and environmental states on integrated dashboards. ○ Parameters: Reactor power (MW), Operator stress index (0–1 scale). ○ KPI: Clarity ratio = $\text{Visible Critical Fields} / \text{Total Critical Fields}$. ● Integrate real-time data streams from simulators, controllers, and sensors. ○ Parameters: Data latency (ms), Number of connected streams. ○ KPI: Data freshness = $1 - (\text{Latency} / \text{Max Threshold})$. ● Guide operators through procedures with adaptive prompts and decision support. ○ Parameters: Procedure step completion time (s), Compliance rate (%). ○ KPI: Procedural adherence index = $\text{Steps completed correctly} / \text{Total steps}$. ● Validate operator actions and plant states against expected behaviors and safety margins. ○ Parameters: Pressurizer pressure (MPa), Boron concentration (ppm). ○ KPI: Validation error = $\text{Measured} - \text{Expected} / \text{Expected}$. ● Monitor system and human performance continuously to detect deviations. ○ Parameters: Heart rate (bpm), Room temperature (°C). ○ KPI: Workload index = $\text{HR deviation from baseline}$. ● Diagnose abnormal conditions by linking alarms to plant variables and human factors. ○ Parameters: Alarm rate (alarms/min), Blink frequency (blinks/min). ○ KPI: Root cause confidence = $\text{Validated anomalies} / \text{Total anomalies detected}$. ● Execute operator and automated commands safely through validated pathways. ○ Parameters: Operator command delay (s), Control rod insertion depth (cm). ○ KPI: Execution accuracy = $\text{Correct commands} / \text{Total commands issued}$. ● Adaptive interface layouts, alarms, and workflows in response to operator workload. ○ Parameters: Ambient light level (lux), Screen refresh frequency of forms. ○ KPI: Adaptation response time = $\text{Time to interface update after trigger}$. ● Train operators in simulated scenarios with integrated performance tracking. ○ Parameters: Scenario completion time (s), Error frequency. ○ KPI: Training improvement = $(\text{Initial error rate} - \text{Current error rate}) / \text{Initial error rate}$. ● Report system and human performance data for feedback, training, and design improvement. ○ Parameters: Number of logs generated, Operator performance index (score). ○ KPI: Reporting completeness = $\text{Logged data points} / \text{Expected data points}$.
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Table A1 shows the main system functions of the digital control room.

Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Nathanael Selvaraj:** Writing – original draft, Visualization, Software, Resources, Formal analysis, Data curation. **Yasir Mahamat:** Writing – review & editing, Software, Resources, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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During the preparation of this work, the author(s) employed the use of AI tools such as ChatGPT to assist in the rearranging and reformatting of several tables and the list of references. The author(s) reviewed, edited and acknowledge the content and take full responsibility for the publication.

Appendix B. Appendix B – System Functions Mapping

Table B1
Main Operation Categories Offered by the Proposed DCR.

● Display is handled mainly by the Digital Control Room, Digital Control Room Model, and human-focused systems like the Human Monitoring System, Human Performance Management System, and Human Performance Semantic Network. These modules ensure plant states, human metrics, and performance insights are visualized clearly. ● Integrate relies on the iPWR Interface, SMR Operation Model, Human Controller, and Control Room Controller. These modules merge data streams from the plant, human operators, and decision support systems into a single control environment. ● Guide is enabled by the SMR Operation Model, HPSN, HPMS, and training tools like the Control Room Simulator. These modules provide adaptive prompts, procedural advice, and context-aware support for operators. ● Validate draws on the iPWR Simulator, SMR Operation Model, HPMS, HMS, and HPSN. Together they confirm that operator actions and system responses match expected behaviors, using real plant dynamics and human performance thresholds. ● Monitor combines data from the iPWR Simulator, HMS, HPMS, and the External Environmental Interface. This function tracks both plant conditions and operator state (physiological, cognitive, behavioral, environmental). ● Diagnose integrates reasoning from the HPSN and metrics from HPMS, with plant context from the iPWR Simulator and SMR Operation Model, to identify abnormal conditions and their likely causes. ● Execute involves the Human Controller, Control Room Controller, and iPWR Interface, ensuring that commands are routed safely from humans or automated systems into plant controls. ● Adapt is powered by HPMS, HPSN, the Digital Control Room, and the Digital Control Room Model, which adjust interface layouts, alarms, and procedures in response to operator workload and cognitive state. ● Train is implemented by the Control Room Simulator, Human Simulator, and iPWR Simulator, with support from HPMS and HMS to provide performance-aware training scenarios. ● Report draws from HPMS, HPSN, the Control Room Simulator, and the Digital Control Room to generate post-scenario reports, trend analyses, and recommendations.
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Table B1 shows the mapping of the main system functions to the different systems of the digital control room.

Appendix C. Appendix C – Design Principles

Table C1
DCR Design Principles.

Layout: Clear, ergonomic design places critical information within easy reach, reducing clutter and operator overload. Modularity: Composed of 12 interconnected modules (e.g., iPWR Simulator, Digital Control Room Model), allowing independent development, easier upgrades, and smooth collaboration. Integration: Seamless data flow between modules ensures operators have up-to-date information and the system can quickly react to changes. Scalability and Adaptability: Easily scales for multiple reactors or training, and adapts displays, alarms, and tools to match operator workload and plant conditions.

Table C1 shows the main design principles of the digital control room. Within this design, these established methodologies were implemented explicitly to how we arranged displays, forms, and fields. Per ISO 11064, important plant parameters and alarms should be presented front center (central, always visible), important secondary information (logs, predictions) were pushed to side displays. Using the Handbook of Control Room Design and Ergonomics, we performed a task analysis of shutdowns, trips, and training sessions to assess what functions should stay together, such as coolant pressure fields and boron concentration during shutdowns. The monitors were validated for their level of access that will be required by operators in a way that was similar to the task-based simulation approaches used in the last two optimization studies with a metro systems control center, by running simulator scenarios using more settled simulation, and determining required status levels to finalize placement for operator reachability, workload, and reaction time level, then changed the placement when it required a faster access to that information. This dynamically produced layout was then actually laid out in proposed initial designs and was robust through the layout from the ergonomics and scenarios without just to concept and theoretical ergonomics and intended based on a task/functional simulation.

Appendix D. Appendix D – Human Controller Specification

The Human Controller connects operator inputs with their physiological and behavioral state, ensuring that commands are interpreted in context. Its core value lies in tracking human condition and adapting how inputs are handled during different operational demands. Main parameters include: Physiological: heart rate, skin temperature, blink rate, facial stress cues; Behavioral: posture, body movement, mouse movement patterns, speech/voice stress; and Task-related: interaction speed, timing of control actions. Scenarios are classified as **Shutdown:** evaluate fatigue and workload during slow, repetitive tasks such as boron injection and power reduction; and **Recovery:** measure stress and reaction time when multiple alarms and quick operator decisions are required. Performance is measured by monitoring physiological signals (heart rate, HRV, skin/core temp, eye-tracking, voice stress). AI models detect stress and fatigue, and workload indices are derived by comparing current values to baseline thresholds. The main functions of the proposed Human Controller module are shown in Fig. D1.

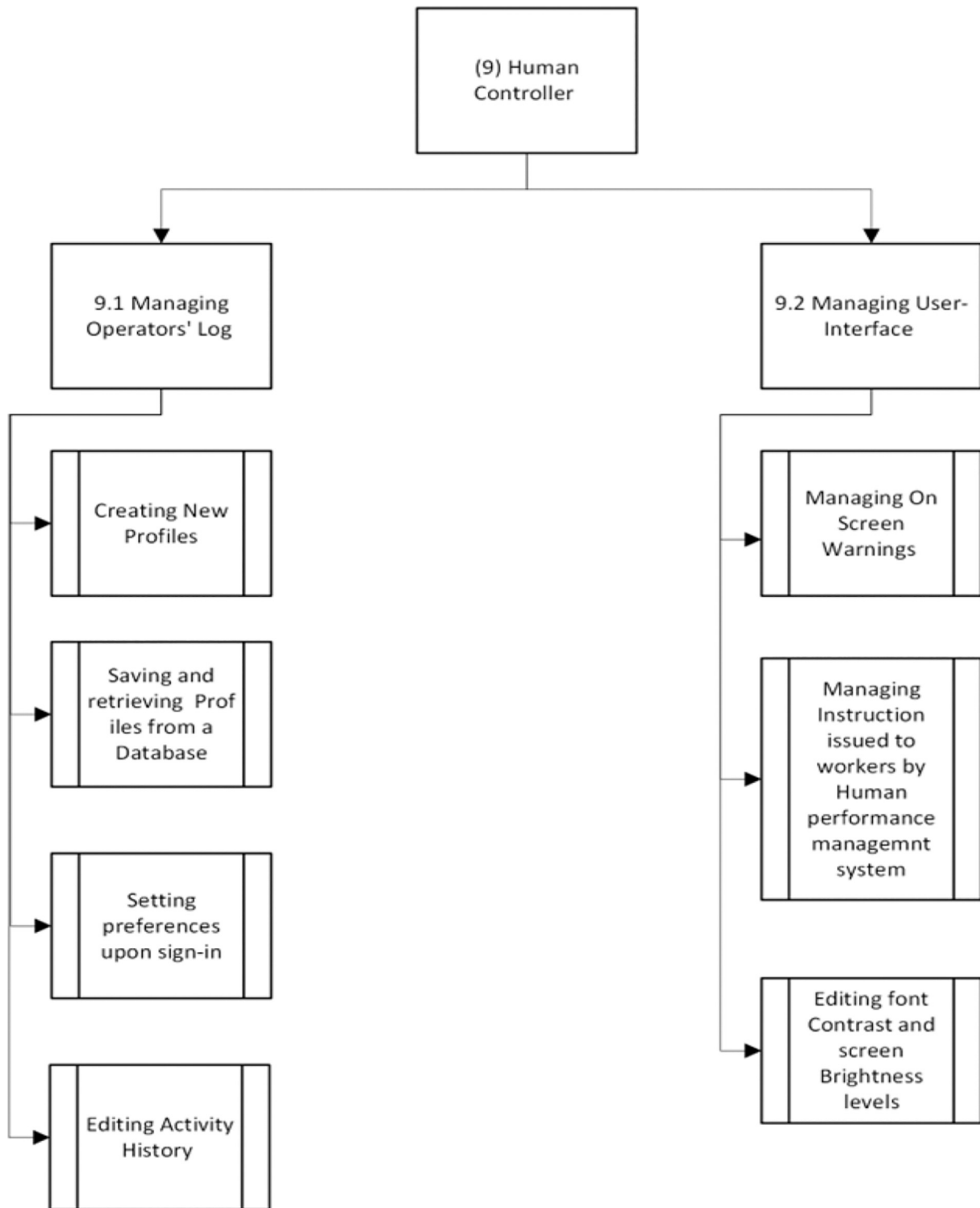


Fig. D1. Human controller functions.

Appendix E. Appendix E – Control Room Controller Specification

The Control Room Controller (CRC) is the main execution and validation engine. It ensures that commands from humans or automation are safe, consistent, and properly logged. Main **parameters are**: Operator inputs: button clicks, slider adjustments, keyboard/mouse actions; Command validation: flags showing approval or rejection of inputs; Plant state data: power level, turbine speed, coolant parameters; and Action logs: timestamps, override frequency, error reports. The relevant scenarios include **Shutdown**: focus on timing of inputs (e.g., ramp-down pacing, boron injection steps); and **Recovery**: analyze speed and accuracy of validation when multiple commands are issued under stress. Performance is measured through system responsiveness and operator-task interactions: alarm acknowledgment times, task completion vs. expected times, and error rates. Efficiency scores and

response delays are compared against predefined standards. The main functions of the proposed CRC are shown in Fig. E1.

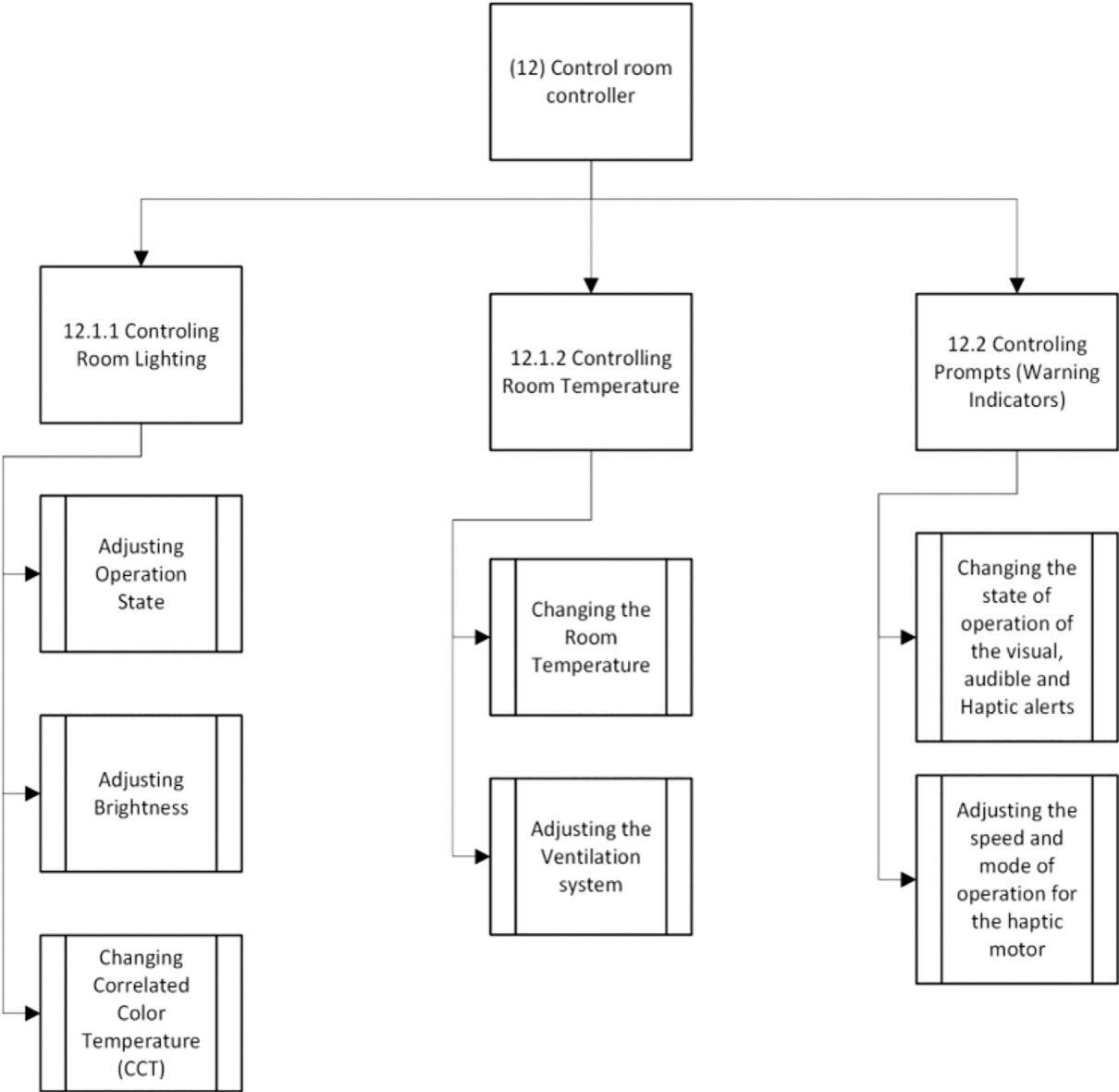


Fig. E1. Control Room Controller Functions.

Appendix F. Appendix F – External Environmental Interface Specification

The External Environmental Interface (EEI) adapts the physical workspace to maintain comfort and reduce strain. It links environmental monitoring directly with human performance data, as shown in Table F1.

Table F1
Environmental Parameters.

Parameters	Description
Room Temperature	Ambient air temperature in the control room; affects comfort, alertness, and cognitive performance.
Light Temperature	Color tone of lighting (Kelvin); influences mood, visual clarity, and cognitive alertness.
Light Intensity	Brightness of lighting (lux); impacts visibility, visual fatigue, and work accuracy.
Humidity	Moisture level in air; low humidity causes dryness/discomfort, high humidity causes fatigue or equipment issues.
Pressure	Atmospheric pressure: deviations may impact comfort, alertness, and equipment performance.

The relevant scenarios include **Shutdown**: examine whether steady lighting and temperature stabilize workload during extended monitoring; and **Recovery**: assess whether adaptive lighting or airflow helps reduce stress spikes and supports faster decision-making. Performance is measured by tracking room conditions (temperature, lighting, noise, humidity) and comparing them to optimal comfort ranges (e.g., 20–24 °C, blue-enriched light for alertness). A comfort score is calculated, and deviations signal reduced performance support. The main functions of the external environment

interface are shown in Fig. F1.

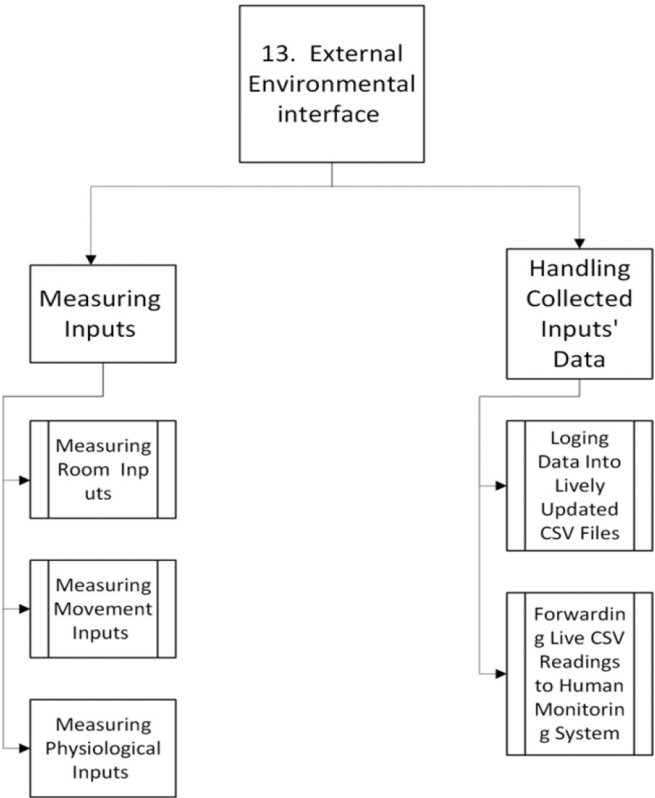


Fig. F1. External Environmental Interface Functions.

Appendix G. Appendix G - Display Layout, Sensor Mapping, and Interface Configuration

The mapping of displays and the corresponding contents are described in Table G1.

Table G1
Mapping of displays & contents.

Display	Default Form	Example Fields	Adaptive Behavior
Display 1	SMR Simulator	Neutron flux, coolant pressure, turbine load	Adds diagnostic strip with alarms during fault conditions; hides low-priority trends under alarm floods.
Display 2	Task Instructions	Current step, next step, timers, validation checks	Enlarges steps and adds hints from Plant Operation GPT (PO-GPT) when operator errors increase or stress is high.
Display 3	Health Monitoring	Heart rate, blink rate, stress index, workload score	Compresses into a strip and moves to Wall Display 3 during high alarm pressure to free Monitor 3 for diagnostics.
Display 4	Plant Overview	KPIs, P&ID summary, mode/state banner	Pins reactor protection status and trip banners at the top during abnormal events.
Display 5	Task Log	Time-stamped operator actions, alarms, outcomes	Switches to instructor annotations in training mode; shows compliance markers in live mode.
Display 6	Alarms & Events	Alarm list with priorities, counts, acknowledgments	Displays compact health strips or predictive trends during calm periods; returns to alarms during stress events.

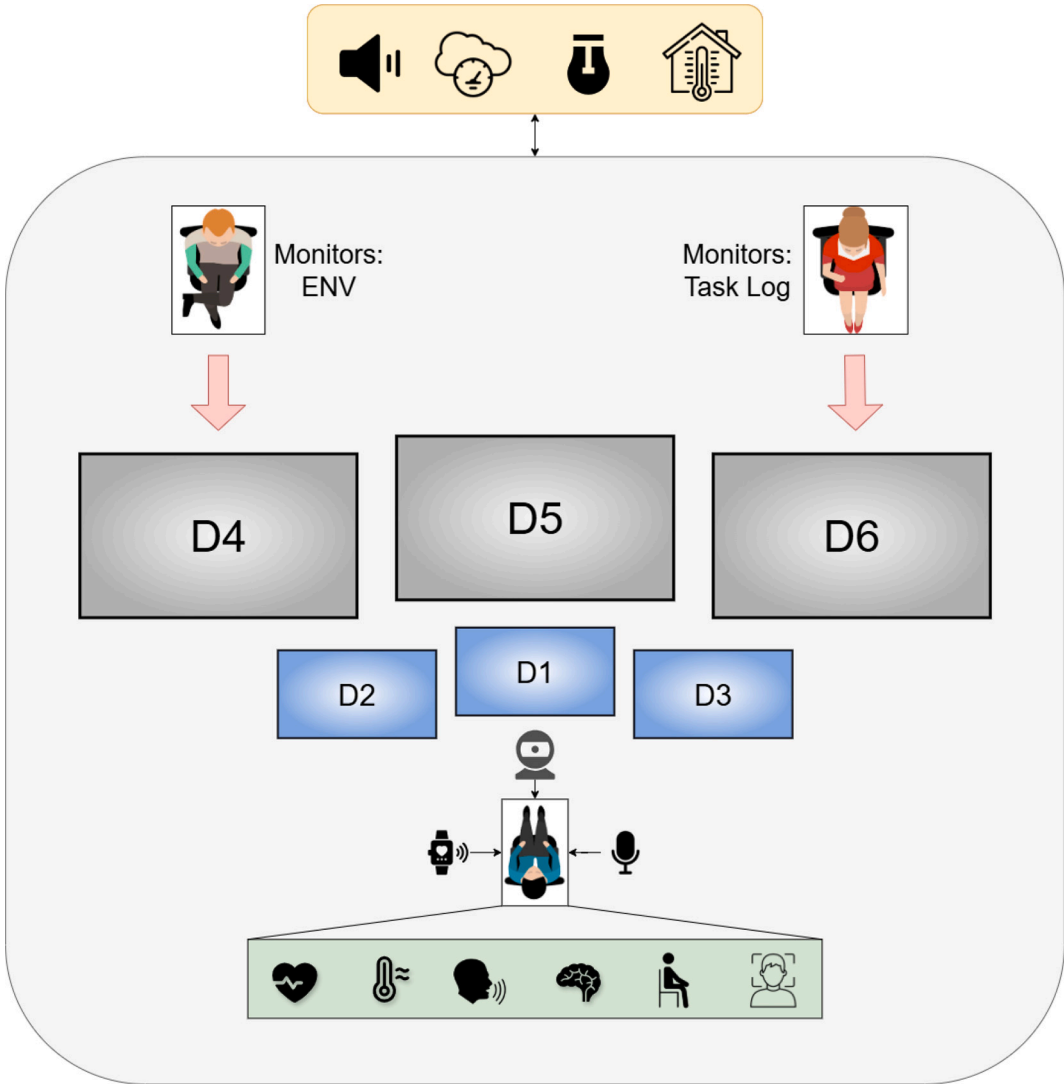


Fig. G1. Digital control room design Layout.

Fig. G1 shows the design layout of the proposed digital control room. The main features of the control room sign are described in Table G2. The overall functions of the digital control room are listed in Table G3.

Table G2
Control room design features.

Feature	Components	Features	Connected Module
Sensors	Microphone	Capture Operator's Speech & Voice Patters	6. HMS
	Thermometer	Captures Room Temperature	13. EEI
	Light Sensor	Captures Room Lighting	13. EEI
	Camera x4	Analyzes Operator's Skin Temperature, Facial emotion, Stress, Eye tracking, Posture & General Movement	6. HMS
	Mouse	Captures Mouse Movement & Speed	6. HMS
Actuators	Smart Watch	Captures Operator's heart rate & blood pressure	6. HMS
	Speakers	Sounds any alerts & alarms	6. HMS & 12. CRC
	Light control	Dims or brightens the room based on conditions	9. HCON
	Temperature Control	Raises or Lowers the ambient room temperature	
	Haptic motor	Vibrates the chair if the operator falls asleep	12. CRC

Table G3
Function mapping into layout.

Function	Where It Appears	Example Content
Display	Large wall displays + monitors 1–3	Simulator trends, procedures, operator health
Integrate	Background system logic	Streams from simulator, sensors, HMS
Guide	Monitor 2	Step-by-step instructions, adaptive prompts
Validate	Monitor 2 + validation highlights	Procedural checkmarks, alarms, interlock warnings
Monitor	Monitor 1 (plant) + Monitor 3 (human)	Plant dynamics, vitals, workload
Diagnose	Wall displays + adaptive alerts	Alarm prioritization, Performance Shaping Factor (PSF) reasoning
Execute	Operator input console	Commands validated and logged
Adapt	All monitors	Dynamic re-ordering of forms, font changes, alarm styling
Train	Simulator scenarios (Monitor 1 + wall displays)	Startup/shutdown exercises
Report	Monitor 3 + exports	After-action performance logs

Appendix H. Appendix H — Detailed Subsystem Metrics for Case Study Scenarios

The different test scenarios and metrics of the HCON, CRC, EEI, iPWR Simulator and ADS are described in [Tables H1 – H5](#).

Table H1
Test Scenario - Human Controller (HCON) Analysis.

Category	Detail
Analyzed Parameters (Average)	Command stream low (2–3 commands/min); all validated on first entry.
Observation	Manual interactions consistent with standard procedure timing.
Interpretation	Operator maintaining normal manual control frequency.
System Response (Action)	Confirms input integrity; logs validated commands; no corrections required.
Reference Metrics	Command Validation Rate = 100%; Input Latency = 40 ms.
Analyzed Parameters (Average)	Manual command rate $\approx$ 3/min; gaze off-center $\approx$ 55% of duration.
Observation	Command input sequence correct but lacks visual confirmation of indicator fields.
Interpretation	Operator relying on memory of expected states (“Turbine Leading,” “Auto”) rather than direct verification.
System Response (Action)	Prompts visual cross-check message on main panel; logs HCON.ALERT.VISCHK.
Reference Metrics	Eye-on-target ratio ≥ 0.7 = sufficient; current 0.45 = under-verification threshold.

Table H2
Test scenario - control room controller (CRC) analysis.

Category	Detail
Analyzed Parameters (Average)	Communication health = 100%; data refresh rate 1 Hz; link stability verified.
Observation	Continuous signal exchange between operator station and iPWR interface without loss.
Interpretation	CRC operating nominally; ready for higher-load phases.
System Response (Action)	Maintains channel synchronization; stores system integrity check record CRC_OK_02.
Reference Metrics	Path Availability = 100%; Round-Trip Latency = 45 ms.
Analyzed Parameters (Average)	Comm health = 99.8%; data refresh 1 Hz; latency = 48 ms; signal integrity verified.
Observation	Stable link despite minor operator attention lapse; system auto-sync maintained.
Interpretation	CRC fully operational; no data lag affecting Rod Control confirmation.
System Response (Action)	Maintains telemetry lock; records CRC_OK_04 snapshot.
Reference Metrics	Path Availability $\geq 99\%$; Latency ≤ 50 ms = Nominal.

Table H3
Test scenario - external environmental interface (EEI) analysis.

Category	Detail
Analyzed Parameters (Average)	Temperature 22 °C, Humidity 46%, Light 520 lx, Noise 32 dB.
Observation	The environment remains within the comfort zone; no operator discomfort signals from HMS.
Interpretation	Stable ambient conditions support optimal operator focus.
System Response (Action)	Continues baseline monitoring; no adjustments triggered.
Reference Metrics	Environmental Comfort Index – ECI = 0.94 (stable).
Analyzed Parameters (Average)	Temp = 23.6 °C, Humidity = 57.7%, Light = 119 lx, Noise = 34 dB.
Observation	Slightly lower illumination compared to previous step; no impact on thermal comfort.
Interpretation	Lighting approaching minimum recommendation; potential contributor to eye fatigue.
System Response (Action)	Sends recommendation to CRC: increase panel illumination to 150 lx.
Reference Metrics	ECI = 0.91 (stable). Illumination Threshold ≥ 120 lx for focus tasks.

Table H4
Test scenario - iPWR simulator / interface analysis.

Category	Detail
Analyzed Parameters (Average)	Reactor power 100%, FW flow 78 kg/s, SG pressure 2.7 MPa, PZR pressure 15.5 MPa, PZR level 43%.Interface latency 45 ms, refresh rate 1 Hz, alarm rate 0/min.
Observation	Full-power steady state confirmed; data transfer and alarm systems nominal.
Interpretation	Plant in stable operating condition; no control deviations detected.
System Response (SIM/IF Action)	Verifies thermal-hydraulic stability; records reference dataset STABLE_100PWR; no adaptive time-scale or validation changes required.
Reference Metrics	Latency = 45 ms; Alarm Rate = 0/min; Command Availability = 100%.
Analyzed Parameters (Average)	Plant Mode = Turbine Leading, Rod Control = Auto, MSB Input = Temp Average; Interface latency 46 ms; refresh 1 Hz; alarm rate 0/min.
Observation	All parameters within expected states; operator interaction present but visual focus low.
Interpretation	Plant control stable, but operator not maintaining full attention on critical readouts — risk of verification error if mode changes occur.
System Response (SIM/IF Action)	Issues reminder dialog: “Confirm Plant Mode Turbine Leading / Rod Auto”; stores state log MODECHK_TL_AUTO.
Reference Metrics	Latency = 46 ms; Command Availability = 100%; Alarm Rate = 0/min → Nominal.

Table H5
Test scenario - adaptive display system (ADS) analysis.

Category	Detail
Analyzed Parameters (Average)	Operator state low stress (SI = 0.25); plant steady (0 alarms); environment 520 lx / 22 °C; no fatigue indicators.
Observation	Operator focus stable across Simulator and Task Instruction forms; no attention drift.
Interpretation	Baseline layout validated; no adaptation needed.
System Response (ADS Action)	Maintains default form mapping (M1 = Simulator, M2 = Instructions, M3 = Health). Confirms layout stability; logs configuration LAYOUT_STABLE_02.
Reference Metrics	Layout Stability Index = 1.00; Visual Comfort Score = 0.95; Refresh Rate = 60 Hz.
Analyzed Parameters (Average)	Operator state: moderate stress (SI = 0.52); focus dispersion = 0.37; plant stable (0 alarms); illumination 119 lx.
Observation	Slight visual overload detected from multi-panel verification (Plant Mode, Rod Control, MSB).
Interpretation	System identifies attention fragmentation; adaptive simplification criteria met.
System Response (ADS Action)	Executes Simplification Algorithm Step 3.2: merge status indicators into single “Plant Status Summary” panel; increase contrast by 10%.
Reference Metrics	Layout Stability Index = 0.82 (after adaptation); Visual Comfort Score = 0.88; Simplification Trigger = (SI > 0.5 ∧ Fixation <0.4).

Appendix I. Appendix I - Performance Measures and Parameters

The defined performance measures with corresponding parameters of IDCR are listed in [Table I1](#).

Table I1
Performance Measures and Parameters of the IDCR.

Performance Measure	Supporting Parameters
Response Time	Time to acknowledge alarms, time to initiate actions, time to complete recovery steps.
Error Rate	Number of procedural mistakes, incorrect control actions, missed validation steps.
Situational Awareness	Eye-tracking patterns, attention distribution, workload scores, operator verbal reports.
Task Efficiency	Duration per procedural step, smoothness of plant recovery, number of corrective actions.
Workload Balance	Heart rate variability, blink frequency, stress indices, posture and movement data.
Training Outcomes	Learning curve progression, consistency across repeated scenarios, resilience under stress.

Data availability

Data will be made available on request.

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